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Formalising frieze patterns in Lean 4

The University of Hong Kong

30 August 2024

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Definitions

Frieze pattern of height n : map

$f : \{0, 1, \dots, n, n + 1\} \times \mathbb{Z} \longrightarrow \mathbb{Z}_{>0}$ with:

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- Border of zeros: $f(0, m) = f(n + 1, m) = 0, \quad \forall m \in \mathbb{Z},$

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$f : \{0, 1, \dots, n, n+1\} \times \mathbb{Z} \longrightarrow \mathbb{Z}_{>0}$ with:

- Border of zeros: $f(0, m) = f(n+1, m) = 0, \quad \forall m \in \mathbb{Z},$
- Border of ones: $f(1, m) = f(n, m) = 1, \quad \forall m \in \mathbb{Z},$

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- Border of zeros: $f(0, m) = f(n+1, m) = 0, \quad \forall m \in \mathbb{Z},$
- Border of ones: $f(1, m) = f(n, m) = 1, \quad \forall m \in \mathbb{Z},$
- Diamond rule: $\forall (i, m) \in \{1, 2, \dots, n\} \times \mathbb{Z},$

$$f(i, m)f(i, m+1) = 1 + f(i+1, m)f(i-1, m+1).$$

Definitions

Frieze pattern of height n : map

$f : \{0, 1, \dots, n, n+1\} \times \mathbb{Z} \longrightarrow \mathbb{Z}_{>0}$ with:

- Border of zeros: $f(0, m) = f(n+1, m) = 0, \quad \forall m \in \mathbb{Z},$
- Border of ones: $f(1, m) = f(n, m) = 1, \quad \forall m \in \mathbb{Z},$
- Diamond rule: $\forall (i, m) \in \{1, 2, \dots, n\} \times \mathbb{Z},$

$$f(i, m)f(i, m+1) = 1 + f(i+1, m)f(i-1, m+1).$$

```
class pattern_n
```

[source](#)

```
  (F : Type u_1) [Field F] (f : ℕ × ℕ → F) (n : ℕ) :
```

```
  Prop
```

```
  topBordZeros : ∀ (m : ℕ), f (0, m) = 0
```

```
  topBordOnes : ∀ (m : ℕ), f (1, m) = 1
```

```
  botBordOnes_n : ∀ (m : ℕ), f (n, m) = 1
```

```
  botBordZeros_n : ∀ (i m : ℕ), i ≥ n + 1 → f (i, m) = 0
```

```
  diamond :
```

```
    ∀ (i m : ℕ),
```

```
      i ≤ n - 1 →
```

```
        f (i + 1, m) * f (i + 1, m + 1) - 1 = f (i + 2, m) * f (i, m + 1)
```

► Instances

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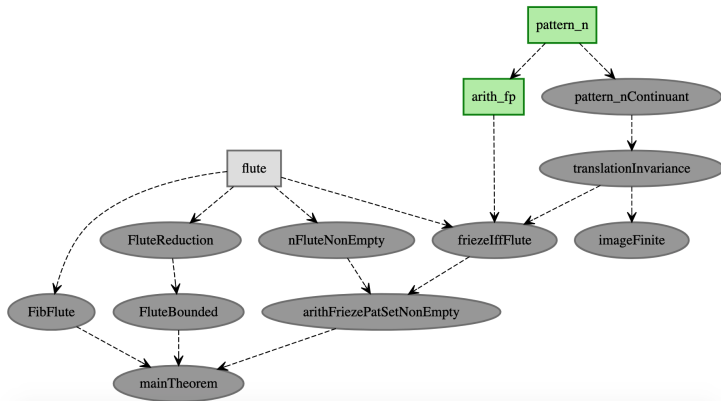
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Translation invariance

Proposition. (Coxeter, 1971) Frieze patterns of height n are $(n + 1)$ -periodic.

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Main theorem

Translation invariance

Proposition. (Coxeter, 1971) Frieze patterns of height n are $(n + 1)$ -periodic.

theorem `translationInvariance`

[source](#)

```
(F : Type u_1) [Field F] (f : ℕ × ℕ → F) (n : ℕ)
[nzPattern_n F f n] (i : ℕ) :
i ≤ n + 1 → ∀ (m : ℕ), f (i, m) = f (i, m + n + 1)
```

Translation invariance

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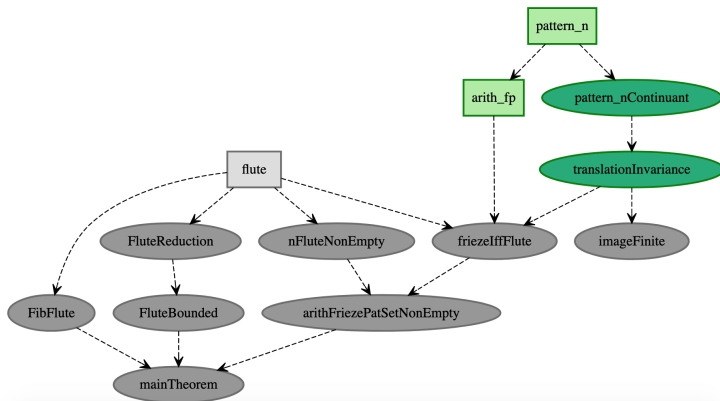
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Definitions

Pan flute of height n : n -tuple of *positive integers* $(a_1, \dots, a_n)$
such that

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Definitions

Pan flute of height n : n -tuple of *positive integers* $(a_1, \dots, a_n)$
such that

- $a_1 = 1$,

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Definitions

Pan flute of height n : n -tuple of *positive integers* $(a_1, \dots, a_n)$
such that

- $a_1 = 1,$
- $a_n = 1,$

Definitions

Pan flute of height n : n -tuple of *positive integers* $(a_1, \dots, a_n)$ such that

- $a_1 = 1$,
- $a_n = 1$,
- $a_i \mid (a_{i-1} + a_{i+1})$, $\forall i \in \{2, \dots, n-1\}$.

```
structure flute
```

[source](#)

```
  (n : N) :
```

```
  Type
```

```
  a : N → N
```

```
  pos : ∀ (i : N), self.a i > 0
```

```
  hd : self.a 0 = 1
```

```
  period : ∀ (k : N), self.a k = self.a (k + (n - 1))
```

```
  div : ∀ (k : N), self.a (k + 1) ∣ self.a k + self.a (k + 2)
```

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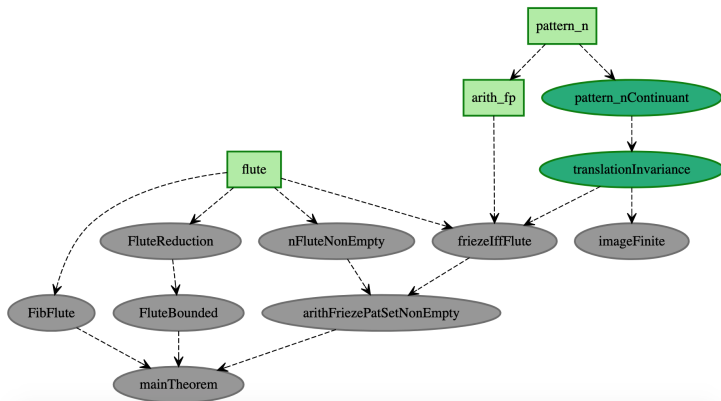
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Fibonacci sequence:

Base. $F_0 = 0, F_1 = 1,$

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Fibonacci sequence:

Base. $F_0 = 0, F_1 = 1,$

IH. $\forall n \geq 0, F_{n+2} = F_{n+1} + F_n.$

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Fibonacci sequence:

Base. $F_0 = 0, F_1 = 1,$

IH. $\forall n \geq 0, F_{n+2} = F_{n+1} + F_n.$

Proposition.

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Fibonacci sequence:

Base. $F_0 = 0, F_1 = 1,$

IH. $\forall n \geq 0, F_{n+2} = F_{n+1} + F_n.$

Proposition. For any flute $(a_1, \dots, a_n)$ of height n , we have

$$a_i \leq F_n.$$

Upper bounds on flutes

Fibonacci sequence:

Base. $F_0 = 0, F_1 = 1,$

IH. $\forall n \geq 0, F_{n+2} = F_{n+1} + F_n.$

Proposition. For any flute $(a_1, \dots, a_n)$ of height n , we have

$$a_i \leq F_n.$$

theorem `FluteBounded`

[source](#)

```
(n : ℕ) (hn : n > 0) (f : flute n) (i : ℕ) :
  i ≤ n - 1 → f.a i ≤ Nat.fib n
```

Upper bounds on flutes

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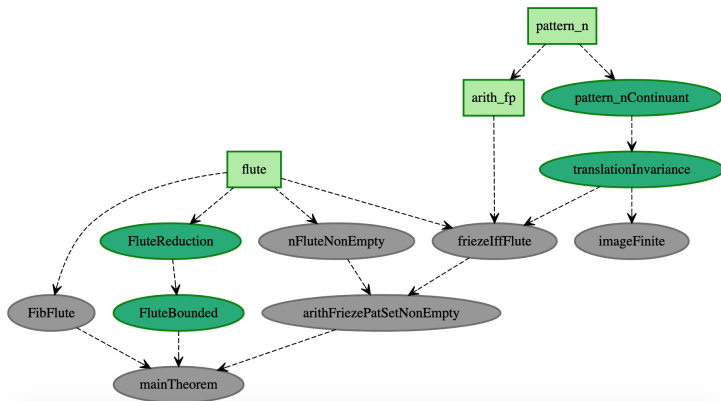
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Proposition. There exists a flute $(f_1, \dots, f_n)$ of height n such that

$$f_i = F_n, \quad \text{for some } i \in \{1, \dots, n\}.$$

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Proposition. There exists a flute $(f_1, \dots, f_n)$ of height n such that

$$f_i = F_n, \quad \text{for some } i \in \{1, \dots, n\}.$$

Proof. By construction.

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Proposition. There exists a flute $(f_1, \dots, f_n)$ of height n such that

$$f_i = F_n, \quad \text{for some } i \in \{1, \dots, n\}.$$

Proof. By construction. When n is even, take

$$(F_1, F_3, \dots, F_{n-1}, F_n, F_{n-2}, \dots, F_4, F_2).$$

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$$f_i = F_n, \quad \text{for some } i \in \{1, \dots, n\}.$$

Proof. By construction. When n is even, take

$$(F_1, F_3, \dots, F_{n-1}, F_n, F_{n-2}, \dots, F_4, F_2).$$

When n is odd, take

$$(F_1, F_3, \dots, F_{n-2}, F_n, F_{n-1}, \dots, F_4, F_2).$$

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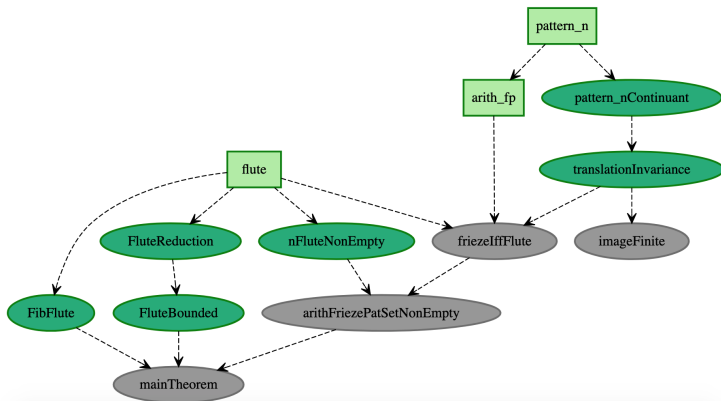
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Friezes consist of flutes

Proposition. Let f be a frieze pattern of height n .

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Main theorem

Friezes consist of flutes

Proposition. Let f be a frieze pattern of height n . For all $m \in \mathbb{Z}$,

$$(f(1, m), f(2, m), \dots, f(n-1, m), f(n, m))$$

is a flute of height n .

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$$(f(1, m), f(2, m), \dots, f(n-1, m), f(n, m))$$

is a flute of height n .

```
def flute_f
```

[source](#)

```
(f : ℕ × ℕ → ℚ) (n : ℕ) (m : ℕ) [arith_fp f n] (i : ℕ) :
```

```
ℕ
```

```
▼ Equations
```

```
• flute_f f n m i = (f (i % (n - 1) + 1, m)).num.toNat
```

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Flutes give rise to friezes

Proposition. Let $f = (f_1, \dots, f_n)$ be a flute of height n .

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Main theorem

Flutes give rise to friezes

Proposition. Let $f = (f_1, \dots, f_n)$ be a flute of height n . There exists a (unique) frieze pattern $\mathcal{F}$ of height n with

$$\mathcal{F}|_{m=0} = (0, f_1, f_2, \dots, f_{n-1}, f_n, 0).$$

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Proposition. Let $f = (f_1, \dots, f_n)$ be a flute of height n . There exists a (unique) frieze pattern $\mathcal{F}$ of height n with

$$\mathcal{F}|_{m=0} = (0, f_1, f_2, \dots, f_{n-1}, f_n, 0).$$

@[irreducible]

[source](#)

def frieze_f

 $\{n : \mathbb{N}\} (g : \text{flute } n) :$ $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{Q}$

▼ Equations

• frieze_f g (i, m) =

if i = 0 then 0

else

if i ≥ n + 1 then 0

else

if m = 0 then ↑(g.a (i - 1)) else

(frieze_f g (i + 1, m - 1) * frieze_f g (i - 1, m) + 1) / frieze_f g (i, m - 1)

Flutes give rise to friezes

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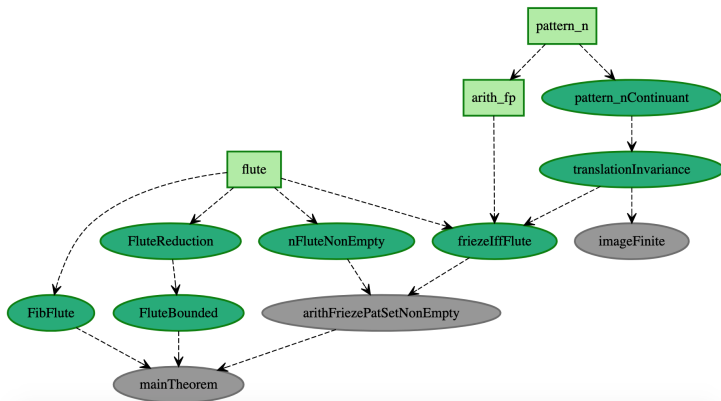
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Main theorem

$$u_n := \max(f(i, m) : \forall i, \forall m, \forall \text{ frieze patterns } f \text{ of height } n).$$

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Main theorem

Main theorem

$u_n := \max(f(i, m) : \forall i, \forall m, \forall \text{ frieze patterns } f \text{ of height } n).$

Theorem. We have

$$u_n = F_n, \quad \forall n \geq 1.$$

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Main theorem

$$u_n := \max(f(i, m) : \forall i, \forall m, \forall \text{ frieze patterns } f \text{ of height } n).$$
Theorem. We have

$$u_n = F_n, \quad \forall n \geq 1.$$

theorem main1[source](#)

```
(n : ℕ) (h : n ≠ 0) (f : ℕ × ℕ → ℚ) :
arith_fp f n → ∀ (a : ℕ × ℕ), f a ≤ ↑(Nat.fib n)
```

theorem main2[source](#)

```
(n : ℕ) (hn : n ≠ 0) :
∃ (f : ℕ × ℕ → ℚ) (h : arith_fp f n) (a : ℕ × ℕ),
f a = ↑(Nat.fib n)
```

theorem main3[source](#)

```
(n : ℕ) (hn : n ≠ 0) (f : ℕ × ℕ → ℚ) (hf : arith_fp f n)
(a : ℕ × ℕ) :
f a ≥ ↑(Nat.fib n) → f a = ↑(Nat.fib n)
```

Our project

What is formalisation

What is Lean 4

Project outline

Part 1: frieze patterns

Definitions

Translation-
invariance

Part 2: Pandean
flutes

Definitions

Upper bounds on
flutes

Fibonacci flutes

Part 3: Proving the
main result

Flutes and friezes

Main theorem

Main theorem

Project complete !